Supplement A

After the Crisis: Outturns, Revisions and Forecast Errors, 2011-2013

Document ID	DOC2
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Edition	March 2014 data vintage, published 15 April 2014
Source classification	Primary statistical supplement
Trust level	High
Relationship to Document 1	Reports outturns for 2011-2013. Supersedes the 2011 and 2012 forecasts (marked * in Document 1, prepared in the fall of 2011) and revises U.S. output growth for 2010.

Teaching document. Prepared for DATA 266 Lab 2 as part of the base knowledge base. Values for 2000-2010 follow Document 1 (Blanchard, *Macroeconomics*, Chapters 1-2) wherever Document 1 reports them, unless a revision is stated explicitly. All other values are constructed for teaching and evaluation and must not be cited as official statistics.

Contents: A.1 World output after the crisis; A.2 The United States, the Euro area and China; A.3 Unemployment: two data vintages; A.4 Stock markets: two base dates; A.5 Unemployment across the Euro area; A.6 The U.S. federal budget; A.7 Vintages, revisions and how to cite this supplement.

Summary

The recovery after 2009 was weaker than forecast in the fall of 2011. World output grew by 2.8% in 2011 and 2.4% in 2012, below the forecasts of 3.0% and 3.2%. The shortfall was largest in the Euro area, where output contracted in both 2012 and 2013 and unemployment kept rising. In the United States, unemployment fell faster than forecast even though output growth stayed close to 2%. China's growth slowed markedly in 2012.

A.1 World output after the crisis

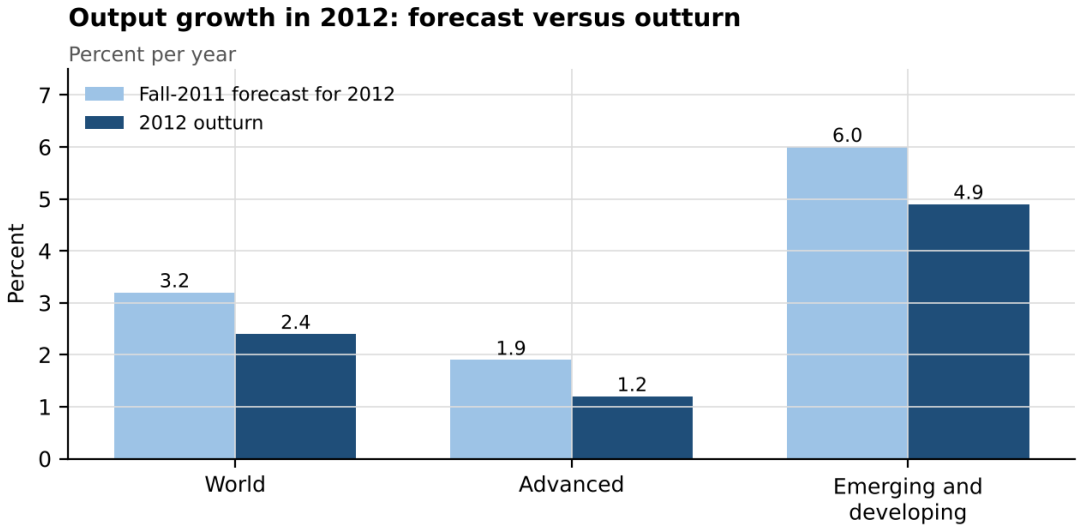
Table A-1 sets the fall-2011 forecasts reported in Table 1-1 of Document 1 against the outturns. Every group grew more slowly than forecast in 2012. The forecast error was largest for emerging and developing economies in absolute terms, although they still grew about four times as fast as advanced economies.

Table A-1 World output growth: forecasts and outturns, 2011-2013

Percent	2011 forecast*	2011 outturn	2012 forecast*	2012 outturn	2013 outturn
World	3.0	2.8	3.2	2.4	2.4
Advanced economies	1.6	1.5	1.9	1.2	1.1
Emerging and developing economies	6.4	6.2	6.0	4.9	4.6

* Forecasts as of the fall of 2011, as reported in Document 1, Table 1-1. Output growth: annual rate of growth of GDP.

Figure A-1 Output growth in 2012: forecast versus outturn



Source: Table A-1.

A.2 The United States, the Euro area and China

Table A-2 reports outturns for the three economies discussed in Document 1. The fall-2011 forecast is shown in brackets after each 2011 and 2012 outturn. Definitions follow Document 1: inflation is the annual rate of change of the GDP deflator, and unemployment is the average over the year.

Table A-2 Growth, unemployment and inflation: outturns 2011-2013 (fall-2011 forecast in brackets)

Percent		2011	2012	2013
United States	Output growth	1.6 (1.5)	2.3 (1.8)	2.2
	Unemployment rate	8.9 (9.1)	8.1 (9.0)	7.4
	Inflation rate	2.1 (2.9)	1.8 (1.2)	1.5
Euro area	Output growth	1.6 (1.6)	-0.7 (1.1)	-0.4
	Unemployment rate	10.2 (9.9)	11.4 (9.9)	12.0
	Inflation rate	1.3 (2.5)	1.3 (1.5)	1.4
China	Output growth	9.3 (9.5)	7.7 (9.0)	7.7
	Unemployment rate	4.1 (4.0)	4.1 (4.0)	4.1
	Inflation rate	7.8 (5.5)	2.0 (3.3)	1.8

Source: DATA 266 Course Data Desk, March 2014 vintage. Forecasts: Document 1, Tables 1-2, 1-3 and 1-4.

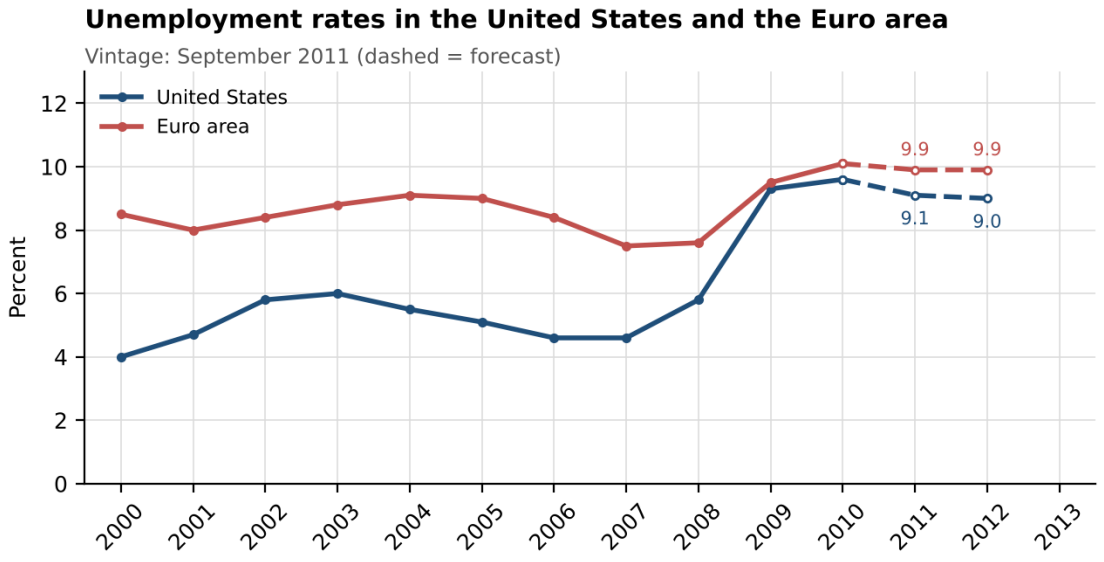
Revision notice. U.S. output growth for 2010 has been revised from 3.0% (Document 1, Table 1-2) to **2.4%** following the comprehensive benchmark revision of the national accounts in July 2013. The revised value supersedes the value in Document 1. No other 2008-2010 value in Document 1 is revised by this supplement.

Three patterns stand out. First, the Euro area went back into recession: output fell in 2012 and 2013, and unemployment rose every year from 2011 to 2013. Second, U.S. unemployment fell by more than the fall-2011 forecast implied, even though output growth was only slightly higher than forecast. Third, China's growth slowed from above 9% to below 8%, while its registered unemployment rate stayed flat.

A.3 Unemployment: two data vintages

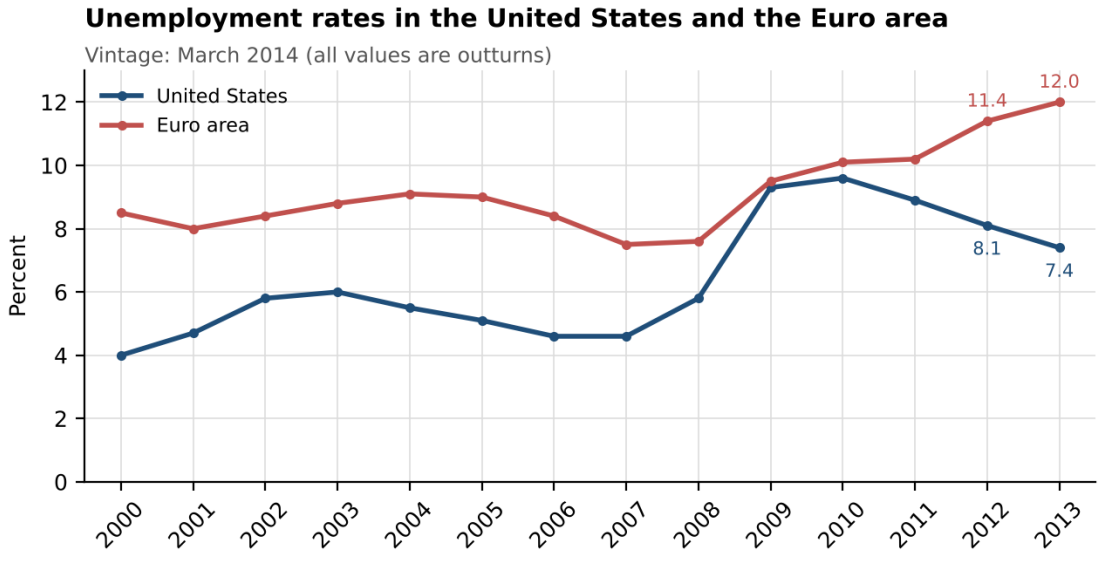
Figures A-2 and A-3 plot the same two unemployment series and look almost identical. They differ in **data vintage**. Figure A-2 reproduces the September 2011 vintage used in Document 1: values for 2011 and 2012 are forecasts and are drawn dashed. Figure A-3 uses the March 2014 vintage: every value is an outturn and the series extends to 2013. Values for 2000-2010 are identical in the two figures. Always state which vintage you are reading.

Figure A-2 Unemployment rates in the United States and the Euro area, 2000-2012 (September 2011 vintage)



Source: Document 1 (World Economic Outlook database, September 2011). Dashed segments are forecasts.

Figure A-3 Unemployment rates in the United States and the Euro area, 2000-2013 (March 2014 vintage)

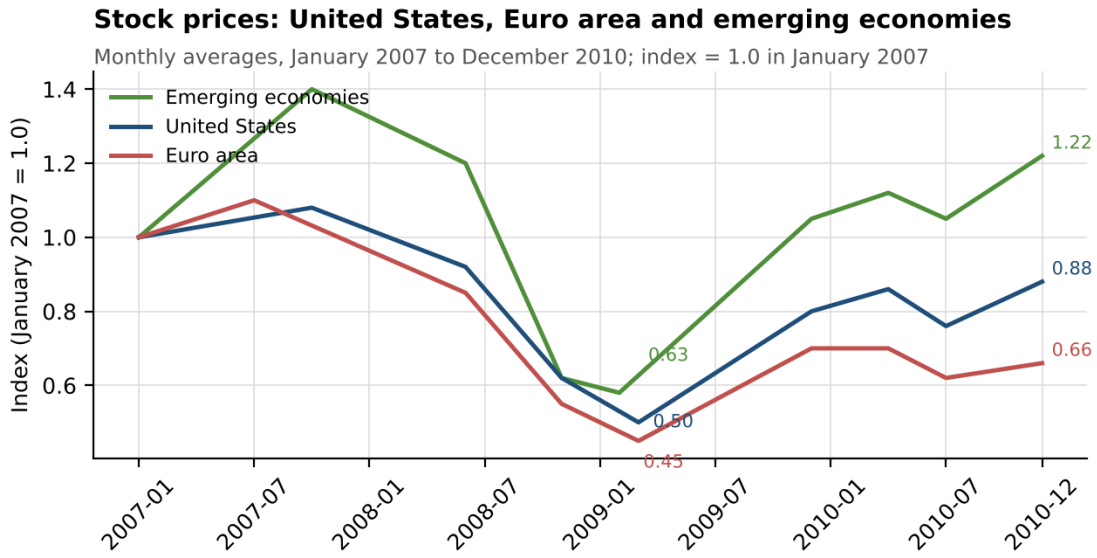


Source: DATA 266 Course Data Desk, March 2014 vintage. All values are outturns.

A.4 Stock markets: two base dates

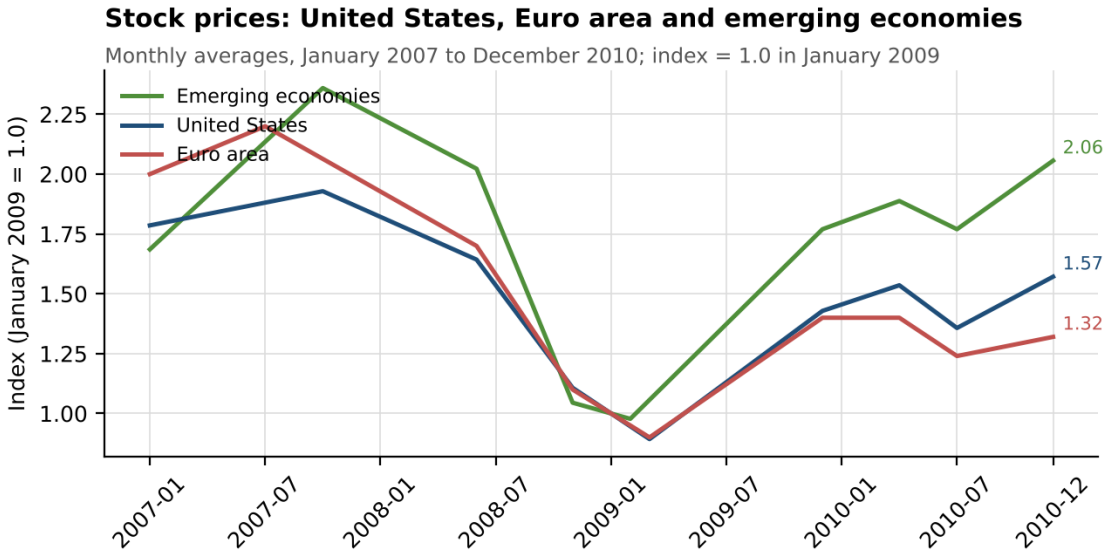
Figures A-4 and A-5 plot the same three monthly stock price indexes over the same period. Figure A-4 sets each index to 1.0 in January 2007, as in Figure 1-1 of Document 1. Figure A-5 divides each series by its January 2009 value, so each index equals 1.0 in January 2009. Rebasing changes every level but not the shape of each line or the percentage change between any two months. Questions about the level of an index must therefore name the base date.

Figure A-4 Stock prices in the United States, the Euro area and emerging economies, 2007-2010 (January 2007 = 1.0)



Source: DATA 266 Course Data Desk; monthly averages.

Figure A-5 Stock prices in the United States, the Euro area and emerging economies, 2007-2010 (January 2009 = 1.0)

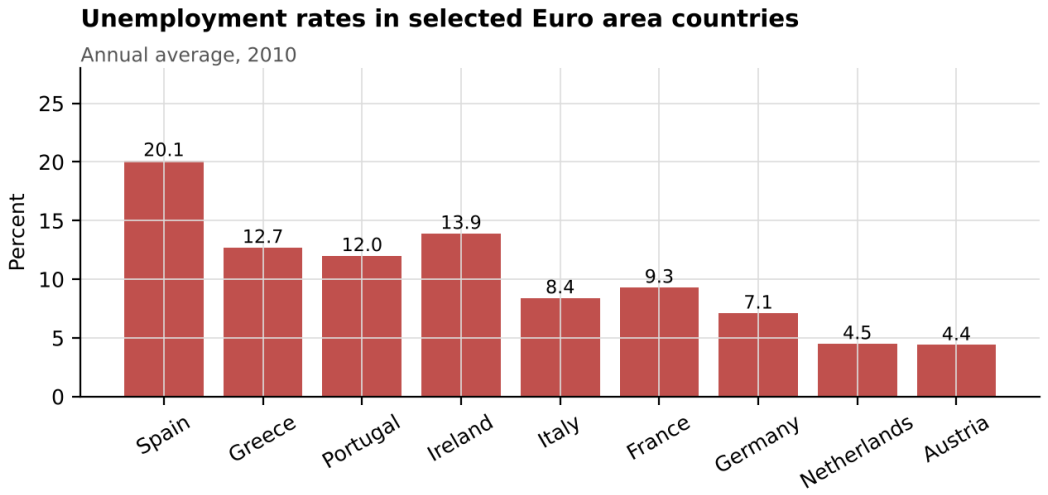


Source: as Figure A-4, rebased to January 2009.

A.5 Unemployment across the Euro area

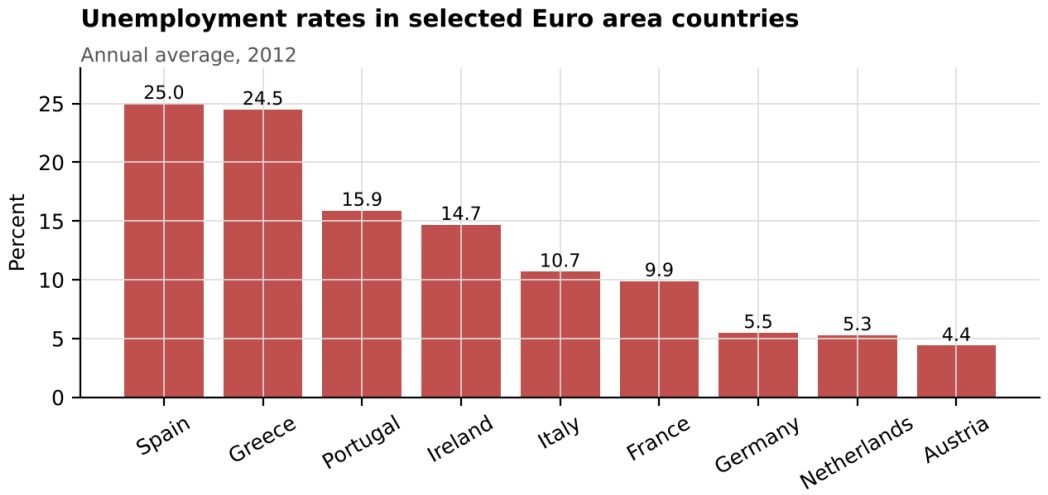
The Euro area average hides very large differences between member countries. Figures A-6 and A-7 show the same nine countries, in the same order and on the same scale, for 2010 and for 2012. Unemployment rose in most of them between the two years, most sharply in the countries at the centre of the sovereign debt crisis. It fell in one of the nine countries and was unchanged in another.

Figure A-6 Unemployment rates in selected Euro area countries, 2010



Source: DATA 266 Course Data Desk; annual averages.

Figure A-7 Unemployment rates in selected Euro area countries, 2012



Source: DATA 266 Course Data Desk; annual averages.

A.6 The U.S. federal budget

Table A-3 extends the budget figures discussed in Document 1. The deficit peaked in 2009 and then narrowed every year, as revenues recovered with output and crisis-related spending was phased out. By 2013 the deficit was back below 5% of GDP, the level Document 1 expected to be reached only "by the middle of the decade".

Table A-3 U.S. federal revenues, spending and budget balance, 2007-2013

Percent of GDP	Revenues	Spending	Budget balance
2007	18.9	20.6	-1.7
2008	17.6	20.8	-3.2
2009	15.1	25.2	-10.1
2010	16.2	25.3	-9.1
2011	15.8	24.5	-8.7
2012	16.4	23.2	-6.8
2013	17.5	21.6	-4.1

A negative budget balance is a deficit. Values for 2007 and 2010 match those quoted in Document 1.

A.7 Vintages, revisions and how to cite this supplement

1. **Outturns supersede forecasts.** Where Document 1 gives a forecast (marked *) for 2011 or 2012 and this supplement gives an outturn for the same year and variable, the outturn is the current value.

2. **Explicit revisions supersede earlier values.** The only revision in this supplement is U.S. output growth for 2010 (see the revision notice in A.2).

3. **Near-identical figures.** Figures A-2/A-3, A-4/A-5 and A-6/A-7 are deliberately drawn with the same template. Cite the exact figure number, and state the vintage, base date or year it refers to.

4. Cite this document as DOC2 with a page number, table or figure, for example "DOC2:Table A-2" or "DOC2:Figure A-3".